CISCO SECURE · HMF DESIGN CLINIC

Team Workbook

Cedarline Health Group — a Hybrid Mesh Firewall design mission

Today your team is the Cisco Solution Engineering team walking into Cedarline. You'll sit with four real conversations, hear what's keeping them up at night, and answer as SEs do: the right solution, why it fits, and the outcome it delivers — pulled together into one Hybrid Mesh Firewall design.

The customer. Cedarline Health Group is a regional healthcare payer-provider — roughly 8,000 employees across four hospitals and thirty-one clinics, plus a claims-processing business that handles the money. Growth by acquisition (including a second data centre absorbed as-is) and a mid-flight, disorderly move to public cloud have left a fragmented security posture. After a security incident in March, the board asked what would have stopped it — and leadership still cannot answer. Budget and executive attention are approved, but explicitly temporary, and the security team is small.

TEAM	MEMBERS	DATE

START HERE

What you'll walk away able to do

1. Map Cedarline's stated concerns to the right Cisco Secure solutions.
2. Explain, in the customer's language, how each solution addresses the concern.
3. Connect the four case studies into one coherent Hybrid Mesh Firewall design.
4. Present the outcome — the business result — not just the feature.

This clinic is built for two kinds of SE

Whether you live and breathe security or you're a generalist SE who's newer to it, this workbook has you covered. Watch for these two markers in every case study:

🧭 **NEW TO THIS AREA?**
Plain-language framing of what the customer is really asking — no security background assumed.

🔐 **FOR SECURITY SPECIALISTS**
The deeper technical angle and the specific Cisco capabilities in play.

How your answers grow — Good, Better, Best

There are no marks and no pass/fail here. Instead, every answer sits somewhere on a growth ladder. Push each one as far up as you can — aim for **Best**.

🌱 Good
NAMES THE FIT

You put the right Cisco solution on the table and show you understand what the customer is actually worried about.

🔷 Better
EXPLAINS THE HOW

You connect that solution to the concern — how it works and why it answers this specific problem — in language the customer would nod along to.

⭐ Best
SELLS THE OUTCOME

You tie it to Cedarline's real constraints and the business outcome they care about, and you place it inside the bigger Hybrid Mesh Firewall story.

The two-part answer. For every customer concern you'll write a **Product / platform** and a **Solution-Engineer response** — how it addresses the concern and the outcome it delivers. That's the SE craft: not "here's a box," but "here's what it changes for you."

The Cisco Secure portfolio, at a glance

A quick decoder so every SE can play — especially if security isn't your day job. (Which product fits which concern is for you to decide.)

Cisco AI Defense	Protects AI apps and models — validation, runtime guardrails, supply-chain scanning, and AI visibility across clouds.
Cisco Multicloud Defense	One security policy plane across AWS, Azure and GCP — segmentation, egress control, and unified visibility.
Cisco Secure Firewall (FTD) + FMC	Next-generation firewall and its manager (Firewall Management Center).
Cisco Security Cloud Control	Cloud-delivered management across the whole firewall estate.
Cisco Secure Workload	Maps how applications actually talk, then drives safe micro-segmentation.
Cisco ISE	Identity Services Engine — knows who and what is on the network, and can quarantine fast.
Isovalent Cilium	eBPF-based Kubernetes networking and policy.
Isovalent Tetragon	eBPF-based runtime security and threat detection for workloads.

Decode the jargon

The terms you'll hear Cedarline use, in plain English.

EU AI Act · Article 15	EU law classifying AI by risk; a 'Provider' must prove robustness and cybersecurity.
MCP (Model Context Protocol)	How AI agents connect to tools and data — a new supply-chain surface to secure.
East-west traffic	Traffic between workloads inside the data centre (vs. north-south, which is in/out).
Lateral movement	An attacker hopping from one compromised system to others on the inside.
Micro-segmentation	Fine-grained isolation of individual workloads so a breach can't spread.
eBPF	Linux kernel technology to observe and enforce networking/security with low overhead.
EVE (Encrypted Visibility Engine)	Infers threats from encrypted traffic patterns without decrypting it.
Snort ML	Machine-learning detection inside the Snort engine for unknown/zero-day exploits.
pxGrid	Cisco's context-sharing fabric so tools can act on identity — e.g. auto-quarantine via ISE.

Hybrid Mesh Firewall (HMF)	Consistent security enforcement across firewalls, cloud gateways, workloads and Kubernetes, managed centrally.

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CASE STUDY 1

AI Security

The conversation. Cedarline is standing up AI across regulated workflows and public cloud faster than security can see it. Help them prove compliance, vet what they adopt, and get visibility and guardrails over AI traffic.

🧭 NEW TO THIS AREA?

AI is now part of the attack surface. Cedarline has to prove to regulators that their AI is safe, vet the AI models and tools they adopt before they go live, and get eyes on AI traffic they currently can't see.

🔐 FOR SECURITY SPECIALISTS

Frame around AI Defense: algorithmic red-teaming for model/app validation, supply-chain scanning of models, repos and MCP servers, AI cloud visibility for asset discovery, and runtime prompt/response guardrails via the AI Defense–Multicloud Defense integration.

PAIN POINT 1 Case Study 1 · AI Security

> *Under the EU AI Act, we are classified as a 'Provider' for our internal AI systems. We need to demonstrate robustness and cybersecurity compliance to satisfy Article 15. How does Cisco help us meet these specific requirements?*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE — HOW IT ADDRESSES THE CONCERN & THE OUTCOME

Good name a fitting Cisco solution & show you get the concern › **Better** explain how it addresses the concern › **Best** tie it to Cedarline's constraints & the outcome, and the wider HMF story.

PAIN POINT 2 Case Study 1 · AI Security

> *We are integrating various open-source models and third-party agents into our internal workflows. How can we ensure these assets aren't introducing vulnerabilities into our environment before they even go live?*

PRODUCT / PLATFORM

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PAIN POINT 3 Case Study 1 · AI Security

> *We know AI is showing up across our cloud environments faster than security can track it. We're worried we have models, agents, datasets, and MCP-connected tools running without visibility or consistent controls.*

PRODUCT / PLATFORM

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> *We're running AI workloads in cloud accounts, but we don't have a reliable way to see which models and AI assets are actually in use. We're also concerned about workloads reaching out to third-party AI models without security knowing about it. And even when we do know the traffic path, we need a practical way to inspect and enforce guardrails on prompt and response traffic in the cloud without changing every application.*

PRODUCT / PLATFORM

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CASE STUDY 2

Cloud Edge

The conversation. Workloads sprawl across AWS, Azure and on-prem, each with its own policy model. Contain lateral movement, prove consistent controls to auditors, and give Cedarline one place to define and enforce policy.

🧭 **NEW TO THIS AREA?**

Cedarline runs applications across several public clouds, each with its own security settings. That inconsistency is risky and painful to audit. They want one place to set policy and prove it.

🔐 **FOR SECURITY SPECIALISTS**

Unified control plane via Multicloud Defense inside HMF; macro/micro-segmentation to contain east-west movement; consolidated logging mapped to frameworks like NIS2/DORA; Secure Workload + Secure Firewall for hybrid policy consistency.

PAIN POINT 1 Case Study 2 · Cloud Edge

> *If a breach occurs in one of our public cloud instances, we are terrified that the attacker will move laterally into our core on-premises systems. How can we contain these threats in a multicloud environment?*

PRODUCT / PLATFORM

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PAIN POINT 2 Case Study 2 · Cloud Edge

> *Our auditors require proof that our security controls are applied consistently across all our cloud workloads. Gathering this data from different cloud providers is a manual, time-consuming nightmare.*

PRODUCT / PLATFORM

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PAIN POINT 3 Case Study 2 · Cloud Edge

> *We have workloads spread across AWS, Azure, and on-premises data centers. Managing security policies individually in each cloud environment is creating massive operational overhead and leading to inconsistent security posture.*

PRODUCT / PLATFORM

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PAIN POINT 4 Case Study 2 · Cloud Edge

> *We have workloads on-prem, in cloud, and in mixed environments, and policy consistency is becoming difficult. Every environment feels different, and we're worried about drift.*

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE — HOW IT ADDRESSES THE CONCERN & THE OUTCOME

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CASE STUDY 3

3 DC Edge · Perimeter Firewall

The conversation. A mixed-vendor perimeter, an inherited rulebase and mostly-encrypted traffic leave blind spots and slow response. Restore visibility, speed the response, and modernise detection with the team they have.

🧭 **NEW TO THIS AREA?**

Their data-centre perimeter is a patchwork of vendors and an inherited rulebase. Most traffic is encrypted, so they're blind to a lot of it — and when something bad happens, the response is too slow.

🔐 **FOR SECURITY SPECIALISTS**

ISE with pxGrid/ANC for rapid, automated quarantine; Encrypted Visibility Engine (EVE) for threat inference without decryption; Snort ML for unknown/zero-day exploits; Security Cloud Control / FMC for policy consolidation and cleanup.

PAIN POINT 1 Case Study 3 · DC Edge · Perimeter Firewall

> *When our security tools detect a compromised device, our manual response time is too slow. By the time we identify the user and manually isolate the port, the attacker has already moved through the network.*

PRODUCT / PLATFORM

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PAIN POINT 2 Case Study 3 · DC Edge · Perimeter Firewall

> *Our firewall environment has become too complex to manage manually.*

PRODUCT / PLATFORM

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PAIN POINT 3 Case Study 3 · DC Edge · Perimeter Firewall

> *Most of our traffic is encrypted, so we've lost visibility into a huge part of our environment.*

PRODUCT / PLATFORM

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PAIN POINT 4 Case Study 3 · DC Edge · Perimeter Firewall

> *We're worried traditional signature-based detection won't catch new or unknown attacks quickly enough.*

PRODUCT / PLATFORM

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CASE STUDY 4

Macro & Micro Segmentation

The conversation. A flat data centre and fragmented Kubernetes networking invite east-west movement. Build segmentation Cedarline can adopt safely — with runtime visibility and application-dependency awareness so nothing breaks.

🧭 **NEW TO THIS AREA?**

The data centre is 'flat' — once an attacker is inside, they can roam. The Kubernetes estate is fragmented too. Cedarline needs segmentation they can actually adopt without breaking applications.

🔐 **FOR SECURITY SPECIALISTS**

Cilium (eBPF) for Kubernetes networking and policy; Tetragon for eBPF runtime security and detection; Cisco Secure Workload for application-dependency mapping so micro-segmentation is designed from real traffic before it is enforced.

PAIN POINT 1 Case Study 4 · Macro & Micro Segmentation

> *Network policy helps, but we also need to know what's happening at runtime inside the workload. We need better detection of suspicious behavior and policy violations.*

PRODUCT / PLATFORM

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PAIN POINT 2 Case Study 4 · Macro & Micro Segmentation

> *Our Kubernetes networking feels fragmented across clusters and environments. We're dealing with too many moving parts, inconsistent policies, and too much operational complexity.*

PRODUCT / PLATFORM

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PAIN POINT 3 Case Study 4 · Macro & Micro Segmentation

> *Right now we're stitching together separate tools for networking, security, and observability in Kubernetes, and it's creating too much overhead.*

PRODUCT / PLATFORM

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PAIN POINT 4 Case Study 4 · Macro & Micro Segmentation

> *We know we need microsegmentation, but we don't have enough visibility into east-west flows or application dependencies to create policy safely. We don't want to break the application.*

PRODUCT / PLATFORM

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Your Hybrid Mesh Firewall design

Four conversations, one story. Combine your case studies into a single HMF design and paste your team's topology below — then be ready to walk Cedarline through it like the trusted advisors you are.

Paste your team's topology diagram here

Show how AI Security, Cloud Edge, DC Edge and Segmentation connect across Cedarline's on-prem, data-centre and multicloud estate.